

A Quick Look at Some of the Best Cloud Platforms for High Performance Computing Applications

Cloud platforms enable high performance computing without the need to purchase the required infrastructure. Cloud services are available on a 'pay per use' basis which is very economical. This article takes a look at cloud platforms like Neptune, BigML, Deep Cognition and Google Colaboratory, all of which can be used for high performance applications.



Software applications, smart devices and gadgets face many performance issues which include load balancing, turnaround time, delay, congestion, Big Data, parallel computations and others. These key issues traditionally consume enormous computational resources and low-configuration computers are not able to work on high performance tasks. The laptops and computers available in the market are designed for personal use; so these systems face numerous performance issues when they are tasked with high performance jobs.

For example, a desktop computer or laptop with a 3GHz processor is able to perform approximately 3 billion computations per second. However,

high performance computing (HPC) is focused on solving complex problems and working on quadrillions or trillions of computations with high speed and maximum accuracy.

Application domains and use cases

High performance computing applications are used in domains where speed and accuracy levels are quite high as compared to those in traditional scenarios, and the cost factor is also very high.

The following are the use cases where high performance implementations are required:

- Nuclear power plants
- Space research organisations
- Oil and gas exploration
- Artificial intelligence and

knowledge discovery

- Machine learning and deep learning
- Financial services and digital forensics
- Geographical and satellite data analytics
- Bio-informatics and molecular sciences

Working with cloud platforms for high performance applications

There are a number of cloud platforms on which high performance computing applications can be launched without users having actual access to the supercomputer. The billing for these cloud services is done on a usage basis and costs less compared to purchasing the actual infrastructure required to work with high performance computing applications.